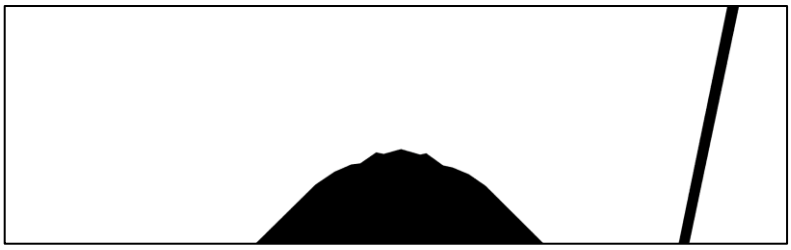
Introduction

In some installations, objects within the Watchkeeper's field of view (FOV), such as antennas or other vessel structures can cause false alarms.
To prevent this, these fixed obstructions must be identified and stored in the system's memory so they are excluded from the AI's detection algorithms.
This process is achieved by creating an **Obstruction Mask**, as explained in the following section.

Obstruction masks are created from the color (RGB) and thermal (IR) camera images. These images are converted into black-and-white duotone representations, where the masked (ignored) areas are marked in black.



This allows the system to distinguish between relevant objects in the scene and static vessel structures, preventing false detections.



The result is an obstruction mask overlaid on the corresponding camera view, visually indicating which areas are excluded from detection..



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REQUIREMENTS

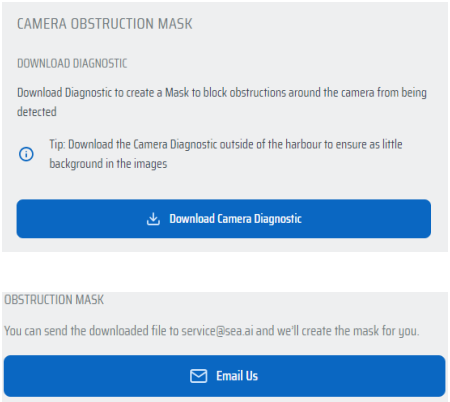
Before creating an ignore mask, ensure the following requirements are met:

1. SEA.AI **Watchkeeper** System
2. **PC** or **Laptop** connected to same network as Watchkeeper
3. **ImageJ Software** – available for download at: https://imagej.net/ij/download.html
4. Watchkeeper Software Version **3.5** or higher
5. **Installer Login Credentials** – password available upon request from service@sea.ai
6. **Operating Environment** – the vessel should be outside of the harbor or positioned where onboard structures are clearly distinguishable from surrounding objects or infrastructure.

DOWNLOADING AND SENDING CAMERA OBSTRUCTION MASK

Before proceeding, ensure that the Watchkeeper system has been properly installed by qualified personnel.

1. On the main User Interface, open the Settings menu.
2. Select the Setup tab and scroll down to the Service & Installation Access login section.
3. In the Password field, enter the service password and select Service Login.
    - If successful, a Service Mode icon will appear on the main User Interface.
4. In the Setup menu, scroll down to the Camera Obstruction Mask section.
5. Select Download Camera Diagnostic.
    - A zipped folder containing two snapshots (Thermal and Color) will be saved to your chosen location.
6. You can optionally send these files to service@sea.ai by clicking the Email Us button.
    - The SEA.AI Professional Services team will create one or both obstruction masks and send them back to you via e-mail.

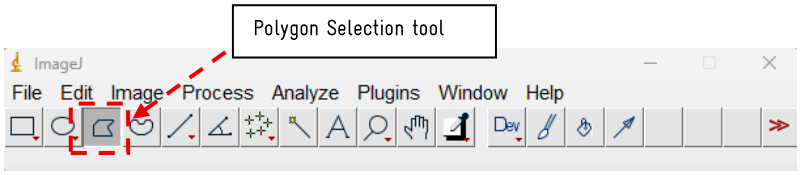


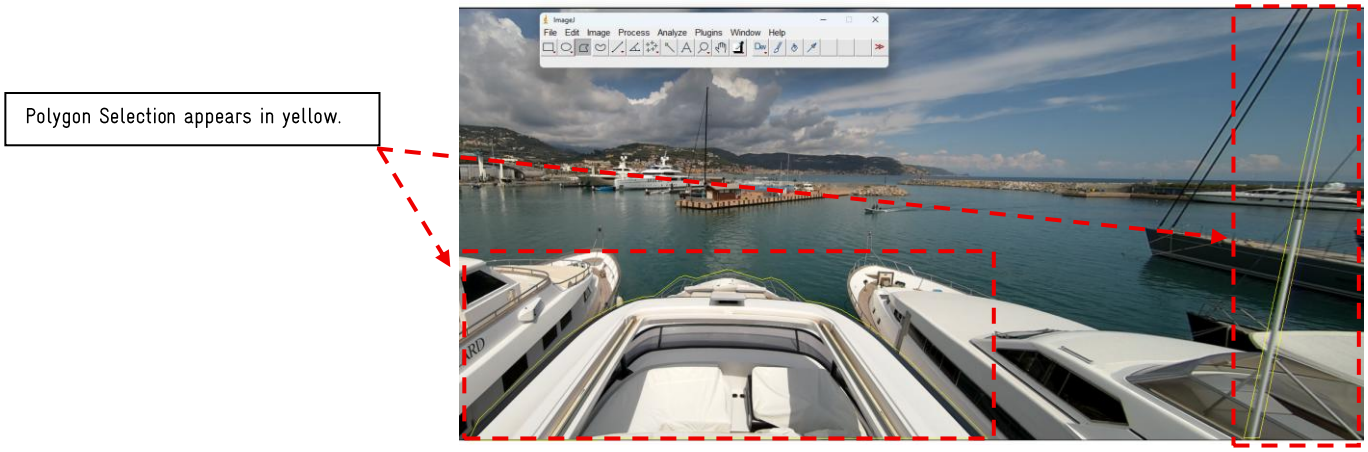
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OBSTRUCTION MASK SELF-CREATION

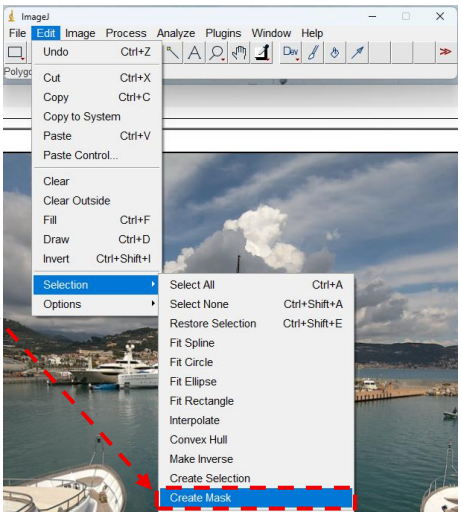
Alternatively, you can create an obstruction mask yourself by following the instructions below:

1. Open ImageJ software.
2. Click File -> Open, then select snapshot file to be modified (Thermal or Color).
3. Select the Polygon Selection tool and outline the vessel structures visible in the field of view.





4. Once all the desired selections are complete, open the **Edit** menu, select the **Selection** submenu, and click **Create Mask**



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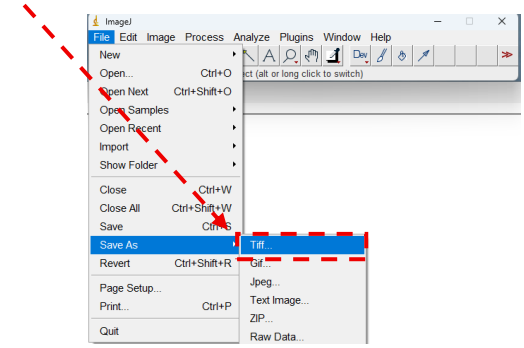
The vessel selection(s) now appears in white. However, AI algorithm excludes parts in black, so the mask must be inverted.



5. Open "Edit" menu and click on "Invert". Now the vessel structural parts are in black.



6. Save the file in ".Tiff" format



7. In the **File Name** field, add the extension **".tiff"** and save the file.
8. The mask must be saved in **TIFF format** for compatibility with the Watchkeeper system.

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UPLOADING AND VERIFYING THE CAMERA OBSTRUCTION MASK

1. In the SEA.AI Watchkeeper User Interface, open the Setup menu and scroll down to the Upload Obstruction Mask section.
    - Field No. 1: Upload the Color Camera Obstruction Mask.
    - Field No. 2: Upload the Thermal Camera Obstruction Mask.
    - Confirm each upload by clicking the corresponding Upload File button.
2. Once the mask(s) have been uploaded, click Show the Camera Mask on the Camera Views to verify that the masks are correctly displayed and aligned.
    - The active mask appears over the camera image (e.g., color obstruction mask shown in red).
    - 

NOTE: An incorrect or poorly aligned mask may cause false detections or reduce performance.
If this occurs, recreate the mask or contact service@sea.ai for assistance.

9. Disable the Show the Camera Mask on the Camera Views function and restart the Watchkeeper system.
10. The setup is now complete.

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